Program #2
Due: Tuesday September 24th, 2019 at 11:59:00 pm

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Purpose

Demonstrate the ability to create and execute a C++ program that utilizes complex data structures to perform searching and sorting.

Assignment

Write a program that will perform sorting and searching across records stored in a file.

The program should first query the user for a filename. The program should open the file and read the first entry. It defines the number of records in the file. The program should read each record and store it into memory. Each record in the file will have sets of three lines. Each set consists of: l*astname, firstname,* and *birthdate*.

Your program should prompt the user for a command:

> *Searching* - The user should be able to supply a search value and your program should print any entries that ***start with*** that value. You should search all first names, last names, and birthdates. There may be multiple values returned.
>
> *Sorting* - The user should be able to choose to print the data in its ***original unsorted form***, sorted by first name, or sorted by last name. Be careful not to lose the original unsorted form.
>
> *Quit* – Exit the program

Deliverables

You must submit your homework through ELearning as a single .zip file. You must include your .cpp files, .h files, and your input file.

- Your project MUST contain multiple .cpp source files
- Each source and header file MUST contain a consistent and full file header (the first lines in the file) that is like that required in program 1.

Notes

No late homework is accepted.

You must use parallel arrays.

You may utilize .find() and .compare() methods.

Input File Format

The input file is a text file that contains a number of records. The first line in the file contains an integer that describes how many records the file contains. After the first line, you will have sets of three lines.

Each set consists of:

Lastname
Firstname
Birthdate

An example file would be:

```
5
Smith
John
01/15/1971
Jones
Ed
06/23/1980
Burns
Sally
03/03/1938
Sinatra
Frank
12/12/1915
Clarkson
Kelly
04/24/1982
```

Recommended Phases

One way to approach the implantation of this assignment is through incremental development. Coding your project in these incremental "phases" may help with the understanding and organization of your code. You are not required to utilize these phases.

Phase 1

Create a working "Hello World" Project

Phase 2

Prompt the user for a file name and then print the file name to the screen.

Phase 3

Attempt to open the input file and then close it. Do not read anything from the file. Put in code that handles both successful and unsuccessful open operations. Only close the file if it was successfully opened. Do not attempt to close a file that failed to open.

Phase 4

Attempt to read the first integer from the file. Print this integer to the screen and verify that you were able to successfully read from the file.

Phase 5

Attempt to read all the entries from the file and print the data to screen. Verify the reads were successful.

Phase 6

Attempt to store all the data from the file into parallel arrays. Once the data is stored in the arrays, traverse each array and print the data to screen to verify you stored it properly and that you can read it back. This

Phase 7

Attempt to implement the search function and print results to screen. This does not re-arrange any data.

Phase 8

Attempt to implement the sort function and print results to screen. This requires rearrangement of data.